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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 9

Mobile Database

Objectives

By the end of this topic you should be able to;

- *Define mobile Positioning*
- *Define mobile databases.*
- *Give some vendors who deal with mobile database*

9.1. Mobile positioning

The terms mobile positioning and mobile location are sometimes used interchangeably in conversation, but they are really two different things.



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- Mobile positioning refers to determining the position of the mobile device.
- Mobile location refers to the location estimate derived from the mobile positioning operation.

There are various means of mobile positioning, which can be divided into two major categories - network based and hand-set based positioning. The purpose of positioning the mobile is to provide location-based services (LBS), including wireless emergency services.

1. Network-based Mobile Positioning Technology

This category is referred to as "network based" because the mobile network, in conjunction with network-based position determination equipment (PDE) is used to position the mobile device.



2. SS7 and Mobile Positioning

One of the easiest means of positioning the mobile user is to leverage the SS7 network to derive location. When a user invokes a service that requires the MSC to launch a message to a LBS residing on a SCP, the MSC may launch a SS7 message containing the cell of origin (COO) or cell ID (of the corresponding cell site currently serving the user). While potentially covering a large area, the COO may be used by LBS to approximate the location of the user. This type of positioning therefore has a large degree of uncertainty that should be taken into account by the LBS application in term of required quality of service (QOS).

- Network based PDE



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COO is not always available nor does it always meet the QOS requirements of the LBS application. Therefore, network-based (or handset based) PDE must be employed.

- Angle of Arrival Method

This method involves analysis of the angle of arrival (AOA) of a signal between the mobile phone and the cellular antenna. AOA PDE is used to capture AOA information to make calculations to determine an estimate of the mobile device position.

- Time of Arrival Method

This method uses the time of arrival (TOA) of signals between the mobile phone and the cellular antenna. TOA PDE is used to capture time difference of arrival (TDOA) information to make calculations to determine an estimate of the mobile device position.



- Radio Propagation Techniques

These techniques utilize a previously determined mapping of the radio frequency (RF) characteristics to determine an estimate of the mobile device position.

- Hybrid Methods

Some hybrid methods of AOA and TOA exist that use the best of both to provide improved positioning.

9.1.1. Handset-based Mobile Positioning Technology

This category is referred to as "handset based" because the handset itself is the primary means of positioning the user, although the network can be used to provide assistance in acquiring the mobile device and/or making position estimate determinations based on measurement data and handset based position deter-



mination algorithms.

- **SIM Toolkit**

The SIM Toolkit (STK), as an API between the Subscriber Identity Module (SIM) of a GSM mobile phone and an application, provides the means of positioning a mobile unit. Positioning information may be as approximate as COO or more precise through additional means such as use of the mobile network operation called timing advance (TA) or a procedure called network measurement report (NMR). In all cases, the STK allows for communication between the SIM (which may contain additional algorithms for positioning) and a location server application (which may contain additional algorithms to assist in mobile positioning). STK is a good technique to obtain position



information while the mobile device is in the idle state.

- **Enhanced Observed Time Difference (E-OTD)**

This is what is also referred to as reversed TOA or handset based TOA. The basic method is employed as with TOA, only the handset is much more actively involved in the positioning process. Specially equipped handsets are required.

9.1.2. GPS

Perhaps the best known or recognized handset based PDE is based on the Global Positioning System (GPS). By itself, GPS can be the most accurate (when satellites are acquired/available), but this technology is often enhanced by the network. Assisted GPS (A-GPS) refers to a PDE system that makes use of addi-



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tional network equipment that is deployed to help acquire the mobile device (much faster than non-assisted GPS) and provide positioning when the A-GPS system is unsuccessful in acquiring any/enough satellites.

9.1.3. Mobile IN Technologies for Positioning

Mobile IN can also be deployed to assist in the positioning process. GSM and ANSI-41 based networks may employ the use of the GSM MAP Any Time Interrogation (ATI) and Position Request (PosReq) messages respectively for positioning.

These mobile IN procedures entail a LBS application middleware as an Service Control Function (SCF) launching a message (MAP ATI or PosReq) to the HLR for position information. The HLR may respond with approximate information (such as



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the COO) or more precise information (such as TA or NMR as in the case with GSM).

The value of mobile IN is to leverage the SS7 and IN network to obtain location, especially for mid-call/session position updates. Mobile IN may also be quite valuable for idle call positioning, but requires integration on the mobile network side to ensure current position information is made available.

9.1.4. The Importance of LBS Middle-ware

LBS middle-ware can be best defined as the applications that do not provide the services themselves, but rather enable location based services.

One of the best examples of LBS middle-ware, that is required for all robust LBS implementations, is the location man-



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ager function. Among other things, the location manager function may be employed to convert positioning information into useful location information and make it available for LBS applications. One of the key value aspects of the location manager function is to enable the use of various positioning technologies in conjunction with various LBS applications. This functional element thus acts as a gateway or hub for location.

Another important role of the location manager function is to perform the SCF function for mobile IN positioning technologies as discussed above. New network entities called the Mobile Positioning Center (MPC) and the Gateway Mobile Location Center (GMLC) are being standardized to provide for the location management function for ANSI-41 and GSM networks respectively.



9.1.5. The Future of Mobile Positioning

There is a bright future for mobile positioning as the key technology for enabling LBS applications, which themselves will become increasingly important as a key enabler of value added services.

Mobile positioning technologies are also crucial for wireless emergency services.

9.2. Mobile Database

A mobile database is a database than can be connected to by a mobile computing device over a mobile network. The client and server have wireless connections. A cache is maintained to hold frequent data and transactions so that they are not lost due to connection failure.



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- Mobile database applications are an effective way to streamline business processes and ensure that end users always have access to the critical corporate information they need to do their jobs.
- Mobile databases let employees enter data on the fly. Information can be synchronized with a server database at a later time.

The use of laptops, mobiles and PDAs is increasing and likely to increase in the future with more and more applications residing in the mobile systems. Many applications such as databases would require the ability to download information from an information repository and operate on this information even when out of range or disconnected. Information is entered on a database but kept in a cache, and then it can be synchronized with a server



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database at a later time.

Mobile applications work best when they include some kind of local data store. By keeping important data local to the device, the user is insulated from network connectivity problems. Data operations are faster (no network lag) and can occur at any time. This results in a much better and less frustrating experience for the user.

"Mobile" is Not the Same as "Online"

Some people equate "mobile" to "online", but they're not the same thing. Online applications require a consistent and dependable network connection to an external server. Browser-based applications are the classic examples of online applications.



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A mobile application is an application that continues to work even when network connectivity is lost, unlike an online application. Being "disconnected" from the network does not stop the application dead in its tracks. This is an important feature for many users.

Many mobile workers move in and out of network coverage on a regular basis while doing their jobs but need information to be "always available". Online solutions don't work well for these kinds of workers.

Depending on the design of the application, keeping track of changes and dealing with potential conflicts can be quite challenging to manage. Mobile database applications often take longer and more skill to develop than their online-only equivalents because of this.



9.2.1. Need for Mobile Databases

- Mobile users must be able to work without a wireless connection due to poor or even non-existent connections.
- Applications must provide significant interactivity.
- Applications must be able to access local device/vehicle hardware, such as printers, bar code scanners, or GPS units (for mapping or Automatic Vehicle Location systems).
- Bandwidth must be conserved (a common requirement on wireless networks that charge per megabyte or data transferred).
- Users don't require access to truly live data, only recently modified data



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If your application meets any of those requirements, the chances are good that you will be required to build a mobile database application with synchronization.

9.3. Mobile Database Architecture

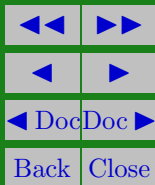
For any mobile architecture, things to be considered are:

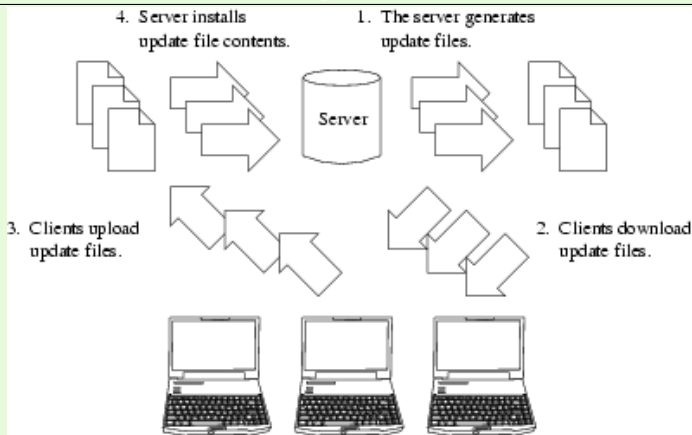
- Users are not attached to a fixed geographical location
- Mobile computing devices: low-power, low-cost, portable
- Wireless networks
- Mobile computing constraints



9.3.1. Architectural Description

Many modern mobile database systems operate while disconnected from the server. In such systems, clients must explicitly synchronize their states. One way to do this is by asynchronously exchanging updates packaged in update files. Clients occasionally package accumulated updates into update files, which are transmitted to a file server when a network connection is established. The server regularly checks the file server for update files, and installs their contents into the database. The server subsequently prepares more update files containing these changes for clients to download. The figure below summarizes the cycle of data through the mobile database architecture.







9.3.2. Problem Description

Among the performance problems with the architecture described above is its inability to scale. In generating the update files for N clients, the server must consider $O(N)$ updates (because the volume of updates is proportional to the number of clients) for each of the N clients. It therefore must do $O(N^2)$ work to generate update files. Second, as the client population grows, so does the load on the file server. As single-server queuing theory suggests, the server will not be able to keep up with demand. In fact, the per-client performance of the file server will decrease with demand. The file server, therefore, cannot scale. Third, the way that update files are installed into the primary database is inefficient. It is currently done on a first-come-first-serve (FCFS) basis. This is inefficient for many reasons. First of all, FCFS



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likely leads to random accesses of the database, which results in low-level inefficiencies. The disk arm has to do many seeks, and the system cache is underutilized. Second of all, many of the updates performed on the database are likely to be undone or overwritten by subsequent operations.

Three parties are involved in Mobile architecture

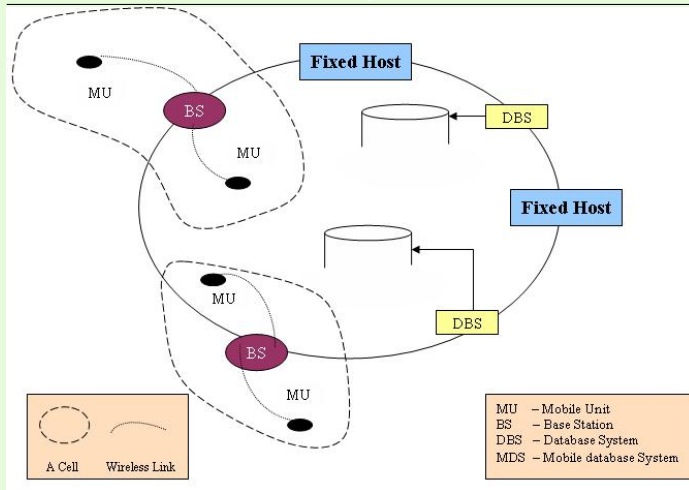
1. **Fixed hosts:** Perform the transaction and data management functions with the help of database servers.
2. **Mobile units:** Portable computers, move around a geographical region that is a collection of mobile cells
 - Mobile hosts retain network connection through the support of base stations
 - Role of mobile hosts depends on the capacity



3. **Base stations:** It is a two-way radio installation in a fixed location, used to communicate with one or more mobile or portable radio transceivers.

9.3.3. Capturing mobility by hand-off processes

- When a mobile unit leaves a mobile cell serviced by a base station, it transfers the responsibility for mobile transaction and data support to the new base station.
- Transparent processes



9.3.4. Vendor's Mobile Databases

With regard to mobile databases: -



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- Sybase Inc.'s SQL Anywhere dominates the field, (about 68% percent of the mobile database market).
- IBM's DB2 Everyplace is a relational database and enterprise synchronization server that extends enterprise applications to mobile devices.
- Microsoft SQL Server Compact and Oracle9i Lite are similar mobile databases.
- There are products from lesser-known vendors, such as SQLBase , Calif., HanDBase from DDH Software Inc, Fla and Database Viewer Plus.

9.3.5. Uses

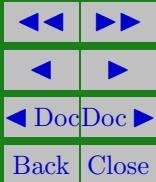
Mobile databases are highly concentrated in Retail and Logistic industry. It is increasingly being used in Aviation and Trans-



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portation Industry





Revision Questions

EXERCISE 1.  Explain why mobile databases are needed

Example . Explain the Mobile file and Database Systems listed below.

Solution:

Wireless File System Access:

Disconnected File Systems:

Mobile Access to C/S or Distributed Databases:

Ad-Hoc Database Systems:

Checkpointing:

Database recovery:

Mobile Database Design:

EXERCISE 2.  Explain the mobile database architecture



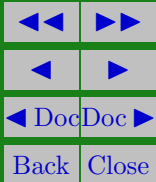
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Solutions to Exercises

Exercise 1. Part of your Revision.

Exercise 1





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Exercise 2. Part of your Revision.

Exercise 2

